transformers are, however, limited by accuracy requirements, rather than heating.

Example 3.7 *A potential transformer with nominal ratio 1100/110 V has the following parameters:*

Primary resistance = 82 Ω secondary resistance = 0.9 Ω
Primary reactance = 76 Ω secondary reactance = 0.72 Ω
No load current = 0.02 a at 0.4 power factor
Calculate (a) phase angle error at no load
(b) burden in VA at unity power factor at which phase angle error will be zero

Solution No load power factor = $\cos(90° + \alpha) = 0.4$

$\therefore\ \alpha = 23.6°$ and $\sin\alpha = 0.4$, $\cos\alpha = 0.917$

$$I_C = I_0 \sin\alpha = 0.02 \times 0.4 = 0.008 \text{ A}$$

$$I_M = I_0 \cos\alpha = 0.02 \times 0.0.917 = 0.0183 \text{ A}$$

Turns ratio $n = 1100/110 = 10$

From Eq. (3.21), $\theta = \dfrac{\dfrac{I_S}{n}(X_P \cos\delta - R_P \sin\delta) + I_C x_P - I_M r_P}{nV_S}$ rad

(a) At no load, $I_S = 0$

$$\therefore \quad \theta = \frac{I_C x_P - I_M r_P}{nV_S} = \frac{0.008 \times 76 - 0.0183 \times 82}{10 \times 100} = -3'06''$$

(b) At unity power factor, $\delta = 0$; thus $\cos\delta = 1$ and $\sin\delta = 0$

From the equation, $\theta = \dfrac{\dfrac{I_S}{n}(X_P \cos\delta - R_P \sin\delta) + I_C x_P - I_M r_P}{nV_S}$ we have

$$\theta = \frac{\dfrac{I_S}{n}(X_P \cos\delta) + I_C x_P - I_M r_P}{nV_S}$$

For zero phase-angle error, $\theta = 0$;

$$\text{Thus,} \quad \theta = \frac{\frac{I_S}{n}(X_P \cos\delta) + I_C x_P - I_M r_P}{nV_S} = 0$$

$$\text{or,} \quad I_S = (I_M r_P - I_C x_P)\frac{n}{X_P}$$

X_P = Total impedance referred to primary = $x_P + n^2 \times X_S$

$$\text{or,} \quad X_P = 76 + 10^2 \times 0.72 = 148\ \Omega$$

$$\text{Thus, or,} \quad I_S = (0.0183 \times 82 - 0.008 \times 76)\frac{10}{148} = 0.06\ \text{A}$$

$\therefore$ secondary burden = $V_S \times I_X = 110 \times 0.06 = 6.6$ VA

Example 3.8 *A potential transformer rated at 6600/110 V has 24000 turns in the primary and 400 turns in the secondary winding. With rated voltage applied to the primary and secondary circuit opened, the primary winding draws a current of 0.004 A, lagging the voltage by 75°. In another operating condition with a certain burden connected to the secondary, the primary draws 0.015 A at an angle 60° lagging with respect to the voltage. The following parameters are given for the transformer:*

Primary resistance = 600 Ω secondary resistance = 0.6 Ω

Primary reactance = 1500 Ω secondary reactance = 0.96 Ω

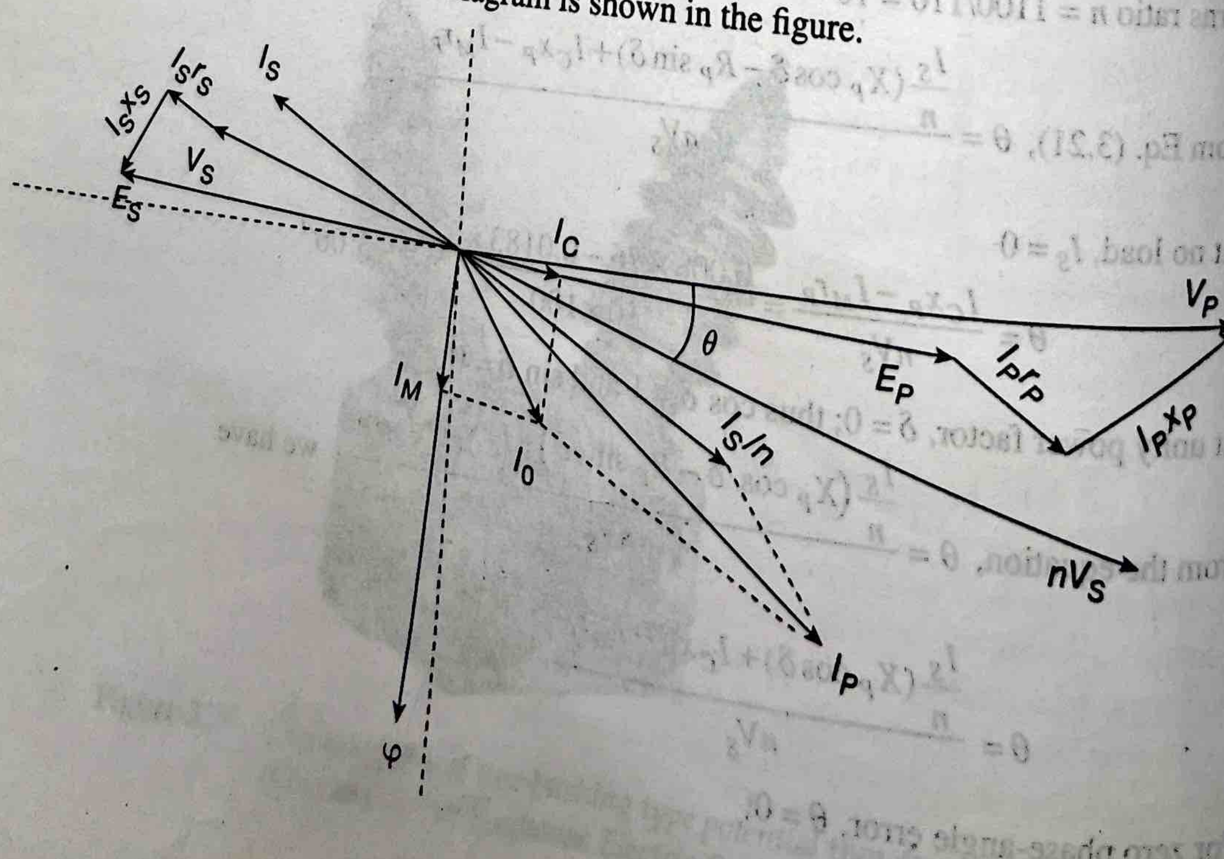
Calculate (a) Secondary load current and terminal voltage, using rated applied voltage as the reference

(b) The load burden in this condition

(c) Actual transformation ratio and phase angle

(d) How many turns should be changed in the primary winding to make the actual ratio equal to the nominal ratio under such operating condition?

Solution The corresponding phasor diagram is shown in the figure.



Nominal ratio = 6600/110 = 60
Turns ratio = 24000/400 = 60
No load current = 0.004 A
No load power factor = cos 75° = 0.26 and sin 75° = 0.966
Primary current = 0.015 A
Primary power factor = cos 60° = 0.5 and sin 60° = 0.866
Primary voltage V_P is taken as reference

Thus, $V_P = 6600 + j0$

$$I_P = 0.015(0.5 - j0.866) = 0.0075 - j0.013$$

$$I_0 = 0.004(0.26 - j0.966) = 0.00104 - j0.0039$$

As seen from Figure 3.21, the phasor $\frac{I_S}{n}$ is the phasor difference of I_P and I_0.

Thus, $\frac{I_S}{n} = (0.0075 - j0.013) - (0.00104 - j0.0039) = 0.00646 - j0.0091$

Thus, I_S (reversed) $= 60 \times (0.00646 - j0.0091) = 0.388 - j0.546$

∴ secondary current phasor $I_S = -(0.388 - j0.546) = -0.388 + j0.546$

(a) Secondary current magnitude $I_S = \sqrt{(0.388)^2 + (0.546)^2} = 0.67$ A

Primary induced voltage

$$E_P = V_P - I_P \times Z_P$$

or, $E_P = 6600 + j0 - (0.0075 - j0.013)(600 + j1500)$

or, $E_P = 6576 - j3.45$

Secondary induced voltage (reversed)

$$E_S(\text{reversed}) = \frac{E_P}{n} = \frac{6576 - j3.45}{60} = 109.6 - j0.0575$$

∴ secondary induced voltage phasor = $E_S = -(109.6 - j0.0575)$

or, $E_S = -109.6 + j0.575$

∴ secondary terminal voltage

$$V_S = E_S - I_S \times Z_S$$

$$= -109.6 + j\,0.0575 - (-0.388 + j0.546)(0.6 + j0.96)$$

$$= -109.9 + j0.76$$

∴ magnitude of secondary terminal voltage

$$V_S = \sqrt{(109.9)^2 + (0.76)^2} = 109.9 \text{ A}$$

(b) Secondary load burden = $V_S \times I_S = 109.9 \times 0.67 = 73.63$ VA

(c) Actual transformation ratio $R = \frac{V_P}{V_S} = \frac{6600}{109.9} = 60.055$

$$V_S \text{ (reversed)} = -(-109.9 + j0.76) = 109.9 - j0.76 \text{ V}$$

$\therefore$ Phase angle by which V_S(reversed) lags V_P

$$\theta = \frac{180}{\pi} \times \tan^{-1}\left(\frac{0.76}{109.9}\right) = 23.7'$$

(d) In order to make actual ratio equal to the nominal ratio, the primary number of turns should be reduced to

$$N'_P = 24000 \times \frac{60}{60.055} \approx 23978$$